Example 5.1 In which quadrant does the following points lie?

(a) (3,–8) (b) (–1,–3) (c) (2, 5) (d) (–7, 3)

Solution

(a) The x- coordinate is positive and y – coordinate is negative. So, point(3,–8) lies in the IV quadrant.

(b) The x-coordinate is negative and y – coordinate is negative. So, point(–1,–3) lies in the III quadrant.

(c) The x-coordinate is positive and y – coordinate is positive. So point(2,5) lies in the I quadrant.

(d) The x-coordinate is negative and y – coordinate is positive. So, point(–7,3) lies in the II quadrant

Example 5.2 Plot the points $A(2,4)$, $B(-3,5)$, $C(-4,-5)$, $D(4,-2)$ in the Cartesian plane.

Solution

(i) To plot (2, 4), draw a vertical line at $x = 2$ and draw a horizontal line at $y = 4$. The intersection of these two lines is the position of (2, 4) in the Cartesian plane. Thus, the Point A (2, 4) is located in the I quadrant of Cartesian plane.

(ii) To plot (–3, 5), draw a vertical line at $x = -3$ and draw a horizontal line at $y = 5$. The intersection of these two lines is the position of (–3, 5) in the Cartesian plane. Thus, the Point B (–3, 5) is located in the II quadrant of Cartesian plane.

(iii) To plot (–4, –5), draw a vertical line at $x = -4$ and draw a horizontal line at $y = -5$. The intersection of these two lines is the position of (-4, 5) in the Cartesian plane. Thus, the Point C (–4, –5) is located in the III quadrant of Cartesian plane.

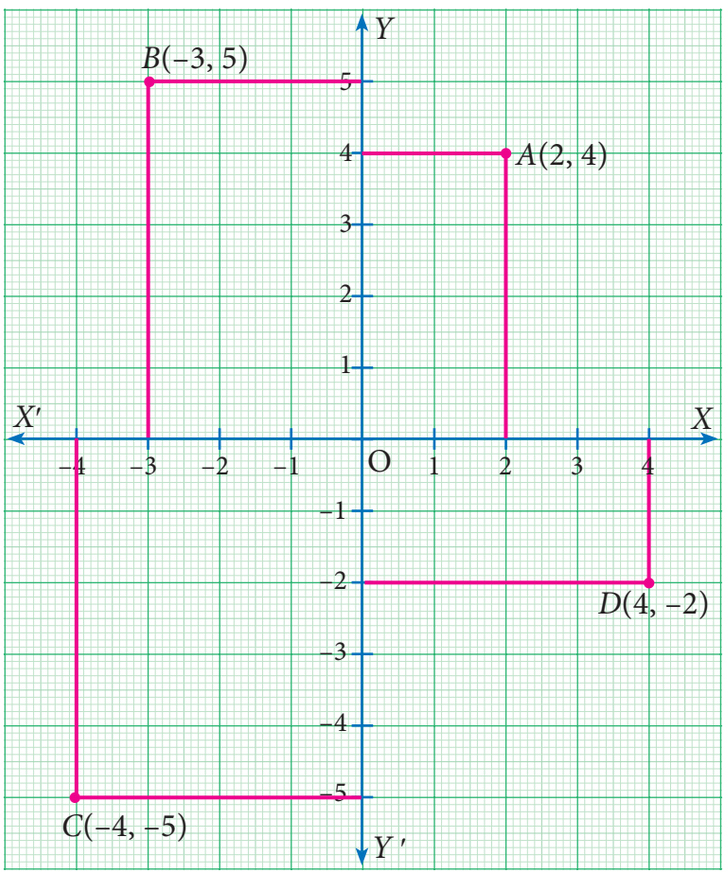


Fig. 5.9

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